

1.4 Exercises

- Let $A_0, \dots, A_n$ be the vertices of a polygon. Determine $\overrightarrow{A_0A_1} + \overrightarrow{A_1A_2} + \dots + \overrightarrow{A_{n-1}A_n} + \overrightarrow{A_nA_0}$.
- In each of the following cases, decide if the indicated vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ can be represented with the vertices of a triangle:
 - $\mathbf{u}(7, 3), \mathbf{v}(2, 8), \mathbf{w}(-5, 5)$.
 - $\mathbf{u}(1, 2, -1), \mathbf{v}(2, -1, 0), \mathbf{w}(1, -3, 1)$.

- Let $ABCD$ be a quadrilateral. Let M, N, P, Q be the midpoints of $[AB], [BC], [CD]$ and $[DA]$ respectively. Show that

$$\overrightarrow{MN} + \overrightarrow{PQ} = \mathbf{0}.$$

Deduce that the midpoints of the sides of an arbitrary quadrilateral form a parallelogram.

- Let $ABCD$ be a quadrilateral. Let E be the midpoint of $[AC]$ and let F be the midpoint of $[BD]$. Show that

$$\overrightarrow{EF} = \frac{1}{2}(\overrightarrow{AB} + \overrightarrow{CD}) = \frac{1}{2}(\overrightarrow{AD} + \overrightarrow{CB}).$$

- Let $ABCD$ be a quadrilateral. Let E be the midpoint of $[AB]$ and let F be the midpoint of $[CD]$. Show that

$$\overrightarrow{EF} = \frac{1}{2}(\overrightarrow{AD} + \overrightarrow{BC}).$$

- Let A', B' and C' be midpoints of the sides of a triangle ABC . Show that for any point O we have

$$\overrightarrow{OA'} + \overrightarrow{OB'} + \overrightarrow{OC'} = \overrightarrow{OA} + \overrightarrow{OB} + \overrightarrow{OC}.$$

- Show that the medians in a triangle intersect in one point and deduce the ratio in which the common intersection point divides the medians.

- In each of the following cases, decide if the given points are collinear:

- | | |
|--------------------------------------|---|
| a) $P(3, -5), Q(-1, 2), R(-5, 9)$. | c) $P(1, 0, -1), Q(0, -1, 2), R(-1, -2, 5)$. |
| b) $A(11, 2), B(1, -3), C(31, 13)$. | d) $A(-1, -1, -4), B(1, 1, 0), C(2, 2, 2)$. |

- Let $ABCD$ be a tetrahedron. Determine the sums

- | | | |
|--|--|--|
| a) $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD}$, | b) $\overrightarrow{AD} + \overrightarrow{BC} + \overrightarrow{DB}$, | c) $\overrightarrow{AB} + \overrightarrow{CD} + \overrightarrow{BC} + \overrightarrow{DA}$. |
|--|--|--|

- Let $ABCD$ be a tetrahedron. Show that

$$\overrightarrow{AD} + \overrightarrow{BC} = \overrightarrow{BD} + \overrightarrow{AC}.$$

11. Let $SABCD$ be a pyramid with apex S and base the parallelogram $ABCD$. Show that

$$\overrightarrow{SA} + \overrightarrow{SB} + \overrightarrow{SC} + \overrightarrow{SD} = 4\overrightarrow{SO}$$

where O is the center of the parallelogram.

12. Give the coordinates of the vertices of the parallelepiped whose faces lie in the coordinate planes and in the planes $x = 1$, $y = 3$ and $z = -2$.

13. In $\mathbb{E}^3$ consider the parallelograms $A_1A_2A_3A_4$ and $B_1B_2B_3B_4$. Show that the midpoints of the segments $[A_1B_1]$, $[A_2B_2]$, $[A_3B_3]$ and $[A_4B_4]$ are the vertices of a parallelogram.

14. Which of the following sets of vectors form a basis?

- a) $\mathbf{v}(1, 2)$, $\mathbf{w}(3, 4)$;
- b) $\mathbf{u}(-1, 2, 1)$, $\mathbf{v}(2, 1, 1)$, $\mathbf{w}(1, 0, -1)$;
- c) $\mathbf{u}(-1, 2, 1)$, $\mathbf{v}(2, 1, 1)$, $\mathbf{w}(0, 5, 3)$;
- d) $\mathbf{v}_1(-1, 2, 1, 0)$, $\mathbf{v}_2(2, 1, 1, 0)$, $\mathbf{v}_3(1, 0, -1, 1)$, $\mathbf{v}_4(1, 0, 0, 1)$;

15. With respect to the basis $\mathcal{B} = (\mathbf{i}, \mathbf{j}, \mathbf{k})$ consider the vectors $\mathbf{u} = \mathbf{i} + \mathbf{j}$, $\mathbf{v} = \mathbf{j} + \mathbf{k}$ and $\mathbf{w} = \mathbf{i} + \mathbf{k}$. Check that $\mathcal{B}' = (\mathbf{u}, \mathbf{v}, \mathbf{w})$ is a basis and give the base change matrix $M_{\mathcal{B}', \mathcal{B}}$.

16. Consider the two coordinate systems $\mathcal{K} = (O, \mathbf{i}, \mathbf{j})$ and $\mathcal{K}' = (O', \mathbf{i}', \mathbf{j}')$ given in Example 1.20. Determine the base change matrix from $\mathcal{K}$ to $\mathcal{K}'$ and the coordinates of the points

$$[A]_{\mathcal{K}} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad [B]_{\mathcal{K}} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}, \quad [C]_{\mathcal{K}} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

in the system $\mathcal{K}'$. Further, determine the base change matrix from $\mathcal{K}'$ to $\mathcal{K}$ and use it with the previously obtained coordinates to calculate $[A]_{\mathcal{K}}$, $[B]_{\mathcal{K}}$ and $[C]_{\mathcal{K}}$.

17. Consider the tetrahedron $ABCD$ and the coordinate systems

$$\mathcal{K}_A = (A, \overrightarrow{AB}, \overrightarrow{AC}, \overrightarrow{AD}), \quad \mathcal{K}'_A = (A, \overrightarrow{AB}, \overrightarrow{AD}, \overrightarrow{AC}), \quad \mathcal{K}_B = (B, \overrightarrow{BA}, \overrightarrow{BC}, \overrightarrow{BD}).$$

Determine

- a) the coordinates of the vertices of the tetrahedron in the three coordinate systems,
- b) the base change matrix from $\mathcal{K}_A$ to $\mathcal{K}'_A$,
- c) the base change matrix from $\mathcal{K}_B$ to $\mathcal{K}_A$.

18. Consider the two coordinate systems $\mathcal{K} = (O, \mathbf{i}, \mathbf{j}, \mathbf{k})$ and $\mathcal{K}' = (O', \mathbf{i}', \mathbf{j}', \mathbf{k}')$ given in Example 1.21. Determine the base change matrix from $\mathcal{K}$ to $\mathcal{K}'$ and the coordinates of the points

$$[A]_{\mathcal{K}} = \begin{bmatrix} 1 \\ 4 \\ -1 \end{bmatrix}, \quad [B]_{\mathcal{K}} = \begin{bmatrix} 1 \\ 5 \\ 1 \end{bmatrix}, \quad [C]_{\mathcal{K}} = \begin{bmatrix} -3 \\ 7 \\ 1 \end{bmatrix}, \quad [D]_{\mathcal{K}} = \begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix}$$

in the coordinate system $\mathcal{K}'$. Further, determine the base change matrix from $\mathcal{K}'$ to $\mathcal{K}$ and use it with the previously determined coordinates to calculate $[A]_{\mathcal{K}}$, $[B]_{\mathcal{K}}$, $[C]_{\mathcal{K}}$ and $[D]_{\mathcal{K}}$.

Midterm : - week 7 : 13.04 $\Rightarrow G_1$
 - week 14 : 08.06 $\Rightarrow G_2$

Final : $0.4 * G_1 + 0.6 * G_2 + \underline{b}$
 $\hookrightarrow b \in [0, 1]$

Minimal conditions:

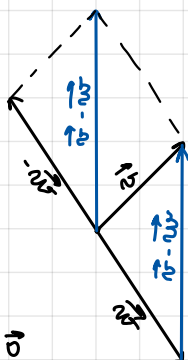
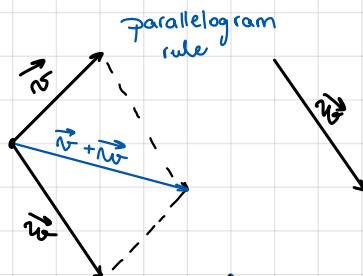
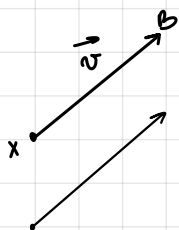
$\hookrightarrow 75\%$ attendance

$\hookrightarrow 0.4 * G_1 + 0.6 * G_2 \geq 5$

Retake: One final exam, $\emptyset$

$\mathbb{E}^n$ n -dimension Euclidean space
 (usually $n = 2$ or 3)
 "space of points"

V^n space of vectors in $\mathbb{E}^n$



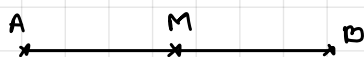
triangle rule

$A_1, A_2, \dots, A_k \in \mathbb{E}^n$

$$\overrightarrow{A_1 A_2} + \overrightarrow{A_2 A_3} + \dots + \overrightarrow{A_{k-1} A_k} + \overrightarrow{A_k A_1} = \vec{0}$$

Fix $O \in \mathbb{E}^n$

$\forall A \in \mathbb{E}^n$ we define $\vec{r}_A = \overrightarrow{OA}$, the position vector of A (with respect to O)



$A, B, M \in \mathbb{E}^n$

M midpoint of $(AB) \Leftrightarrow \vec{r}_M = \frac{\vec{r}_A + \vec{r}_B}{2}$

1.4 ABCD - quadrilateral

E midpoint of [AC]

F midpoint of [BD]

$$\rightarrow \vec{EF} = \frac{1}{2}(\vec{AB} + \vec{CD}) - \frac{1}{2}(\vec{AD} + \vec{CB})$$

$$\vec{EF} = \vec{EO} + \vec{OF} = \vec{r}_F - \vec{r}_E$$

$$\vec{r}_F = \frac{\vec{r}_A + \vec{r}_C}{2}, \quad \vec{r}_E = \frac{\vec{r}_B + \vec{r}_D}{2}$$

$$\frac{1}{2}(\vec{r}_A + \vec{r}_C - \vec{r}_B - \vec{r}_D) = \frac{1}{2}(\vec{AB} + \vec{CD}) - \frac{1}{2}(\vec{AD} + \vec{CB})$$

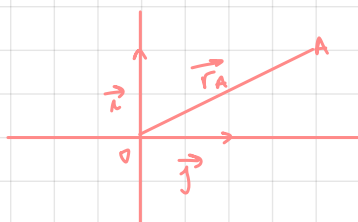
$A, B, C \in E^n$ collinear $\Leftrightarrow \exists$ line, $l \in E^n$ st. $l \ni A, B, C \Leftrightarrow \text{rank}(\vec{AB}, \vec{BC}, \vec{CA}) = 1$
 $\Leftrightarrow \exists \alpha \in \mathbb{R} : \vec{AB} = \alpha \vec{BC}$

$\mathcal{H} = (O, \mathcal{B})$

reference system

- $O \in E^n$

- $\mathcal{B}$ basis of V^n



$$[A]_{\mathcal{H}} = [\vec{r}_A]_{\mathcal{B}}$$

A, B, C collinear

$\Leftrightarrow \exists \alpha \in \mathbb{R}$

$$[\vec{AB}]_{\mathcal{H}} = \alpha [\vec{BC}]_{\mathcal{B}}$$

1.8 Decide if the following points are collinear:

a) $P(3, -5)$, $Q(-1, 2)$, $R(-5, 9)$

$\vec{v}, \vec{w}$ are collinear if $\vec{v} = k\vec{w}$, $k \in \mathbb{R}$

$$\vec{PQ} = \vec{r}_Q - \vec{r}_P = (x_Q - x_P)\vec{i} + (y_Q - y_P)\vec{j}$$

$$= (-1-3)\vec{i} + (2+5)\vec{j}$$

$$= -4\vec{i} + 7\vec{j}$$

$$\vec{QR} = \vec{r}_R - \vec{r}_Q = (x_R - x_Q)\vec{i} + (y_R - y_Q)\vec{j}$$

$$= (-5+1)\vec{i} + (9-2)\vec{j}$$

$$= -4\vec{i} + 7\vec{j}$$

$$\vec{PQ} = \vec{QR} \Rightarrow P, Q, R \text{ are collinear}$$

b) $A(11, 2), B(1, -3), C(31, 13)$

$$\begin{aligned}\vec{AB} &= \vec{r}_B - \vec{r}_A = (x_B - x_A)\vec{i} + (y_B - y_A)\vec{j} \\ &= (1 - 11)\vec{i} + (-3 - 2)\vec{j} \\ &= -10\vec{i} - 5\vec{j}\end{aligned}$$

$$\begin{aligned}\vec{BC} &= \vec{r}_C - \vec{r}_B = (x_C - x_B)\vec{i} + (y_C - y_B)\vec{j} \\ &= (31 - 11)\vec{i} + (13 + 3)\vec{j} \\ &= 20\vec{i} + 16\vec{j}\end{aligned}$$

A, B, C are not collinear

c) $A(1, 0, -1), B(0, -1, 2), C(-1, -2, 5)$

$$\vec{AB} = \vec{r}_B - \vec{r}_A = -\vec{i} - \vec{j} + 3\vec{k}$$

$$\vec{BC} = \vec{r}_C - \vec{r}_B = -\vec{i} - \vec{j} + 3\vec{k}$$

A, B, C collinear

d) $A(-1, -1, 4), B(1, 1, 0), C(2, 2, 5)$

$$\vec{AB} = (0, 0, -4)$$

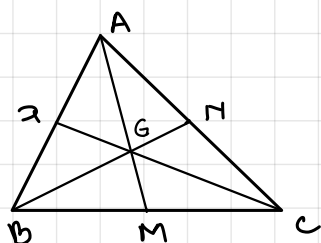
$$\vec{BC} = (1, 1, 5)$$

$$0 \neq 0 \neq \frac{-4}{5} \Rightarrow A, B, C \text{ are not collinear}$$

$\overset{x}{\text{A}} \text{-----} \overset{x}{\text{B}}$
 $A, B, T \in \mathbb{E}^n$

$$T \in AB \Leftrightarrow \exists \lambda \in \mathbb{R} \cdot \vec{r}_T = \lambda \vec{r}_A + (1 - \lambda) \vec{r}_B \quad (\text{vector equation for the line } AB)$$

1.7 Show that the medians in a $\triangle ABC$ intersect in a point, and find the ratio that this point divides the medians in.



M, N, P midpoints of $[BC], [AC], [AB]$

$$\text{Let } \{G\} = AM \cap BN$$

We will show that $G \in CP$

$$G \in AM \Rightarrow \exists \lambda \in \mathbb{R}$$

$$\vec{r}_G = \lambda \vec{r}_A + (1-\lambda) \vec{r}_M = \lambda \vec{r}_A + \frac{1-\lambda}{2} \vec{r}_B + \frac{1-\lambda}{2} \vec{r}_C$$

$$G \in BN \Rightarrow \exists \mu \in \mathbb{R}$$

$$\vec{r}_G = \mu \vec{r}_B + (1-\mu) \vec{r}_N = \frac{1-\mu}{2} \vec{r}_A + \mu \vec{r}_B + \frac{1-\mu}{2} \vec{r}_C$$

$$\vec{r}_A \left(\lambda - \frac{1-\mu}{2} \right) + \vec{r}_B \left(\frac{1-\lambda}{2} - \mu \right) + \vec{r}_C \left(\frac{1-\lambda}{2} - \frac{1-\mu}{2} \right) = \vec{0}$$

$\triangle ABC$ non-degenerate

So, $\vec{v} = \vec{AB}$, $\vec{w} = \vec{AC}$ are linearly independent

$$\vec{r}_B = \vec{r}_A + \vec{v}, \quad \vec{r}_C = \vec{r}_A + \vec{w}$$

$$\vec{r}_A \left(\lambda - \frac{1-\mu}{2} \right) + (\vec{r}_A + \vec{v}) \left(\frac{1-\lambda}{2} - \mu \right) + (\vec{r}_A + \vec{w}) \left(\frac{1-\lambda}{2} - \frac{1-\mu}{2} \right) = \vec{0}$$

$$\vec{r}_A \left(\lambda - \frac{1-\mu}{2} + \frac{1-\lambda}{2} - \mu + \frac{1-\lambda}{2} - \frac{1-\mu}{2} \right) + \vec{v} \left(\frac{1-\lambda}{2} - \mu \right) + \vec{w} \left(\frac{1-\lambda}{2} - \frac{1-\mu}{2} \right) = \vec{0}$$

$\vec{v}, \vec{w}$ lin. indep.

$$\Rightarrow \left\{ \begin{array}{l} \frac{1-\lambda}{2} = \mu \\ \frac{1-\lambda}{2} = \frac{1-\mu}{2} \Rightarrow \lambda = \mu \end{array} \right\} \Rightarrow \frac{1-\lambda}{2} = \lambda \Rightarrow \lambda = \frac{1}{3}$$

$$\Rightarrow \mu = \lambda = \frac{1}{3}$$

$$\Rightarrow \vec{r}_G = \frac{1}{3} \vec{r}_A + \frac{1}{3} \vec{r}_B + \frac{1}{3} \vec{r}_C$$

$$\vec{CG} = \vec{r}_G - \vec{r}_C = \frac{1}{3} \vec{r}_A + \frac{1}{3} \vec{r}_B - \frac{2}{3} \vec{r}_C$$

$$\vec{CP} = \vec{r}_P - \vec{r}_C = \frac{1}{2} \vec{r}_A + \frac{1}{2} \vec{r}_B - \vec{r}_C$$

$$\Rightarrow \frac{2}{3} \vec{CP} = \vec{CG} \Rightarrow C, G, P \text{ collinear and } \frac{CG}{CP} = \frac{2}{3}$$

Me practicing :

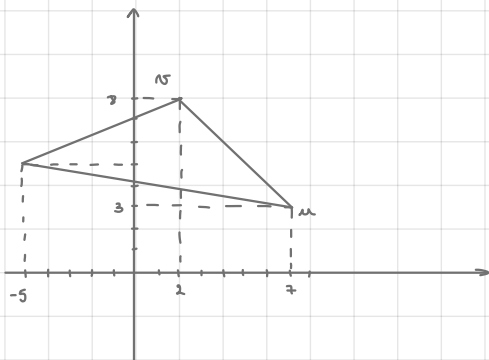
1. $A_0 \dots A_n$ be the vertices of a polygon.

$$\overrightarrow{A_0 A_1} + \overrightarrow{A_1 A_2} + \overrightarrow{A_2 A_3} + \dots + \overrightarrow{A_{n-1} A_n} + \overrightarrow{A_n A_0}$$

$$= \cancel{\vec{r}_1 - \vec{r}_0} + \cancel{\vec{r}_2 - \vec{r}_1} + \cancel{\vec{r}_3 - \vec{r}_2} + \dots + \cancel{\vec{r}_{n-1} - \vec{r}_{n-2}} + \cancel{\vec{r}_n - \vec{r}_{n-1}} + \vec{r}_0 - \vec{r}_n$$

$$= \vec{0}$$

2. a) $u(7,3)$, $v(2,8)$, $w(-5,5)$



if they are not collinear, they can form a triangle

$$\overrightarrow{uv} = \vec{r}_v - \vec{r}_u = -5\vec{i} + 5\vec{j}$$

$$\overrightarrow{vw} = \vec{r}_w - \vec{r}_v = 7\vec{i} - 3\vec{j}$$

b) $u(1,2,-1)$, $v(2,-1,0)$, $w(1,-3,1)$

$$\overrightarrow{uv} = \vec{r}_v - \vec{r}_u = \vec{i} - 3\vec{j} + \vec{k}$$

$$\overrightarrow{vw} = \vec{r}_w - \vec{r}_v = -\vec{i} - 2\vec{j} + \vec{k}$$

} they are not collinear, so they can form a Δ

3. ABCD - quadrilateral, M, N, P, Q midpoints of [AB], [BC], [CD] and [DA]

$$\overrightarrow{MN} + \overrightarrow{PQ} \stackrel{?}{=} \vec{0}$$

2.5 Exercises

2.1. Determine parametric equations for the line $\ell \subseteq \mathbb{A}^2$ in the following cases:

- ℓ contains the point $A(1, 2)$ and is parallel to the vector $\mathbf{a}(3, -1)$,
- ℓ contains the origin and is parallel to $\mathbf{b}(4, 5)$,
- ℓ contains the point $M(1, 7)$ and is parallel to Oy ,
- ℓ contains the points $M(2, 4)$ and $N(2, -5)$.

2.2. For the lines ℓ in the previous exercise

- determine a Cartesian equation for ℓ ,
- describe all direction vectors for ℓ .

2.3. With the assumptions in Example 1.20, give parametric equations and Cartesian equations for the lines AB, AC, BC both in the coordinate system $\mathcal{K}$ and in the coordinate system $\mathcal{K}'$.

2.4. Find a Cartesian equation of the line ℓ in $\mathbb{A}^2$ containing the points $P = S \cap S'$ and $Q = T \cap T'$ where

$$S : x + 5y - 8 = 0, \quad S' : 3x + 6 = 0, \quad T : 5x - \frac{1}{2}y = 1, \quad T' : x - y = 5.$$

2.5. Determine an equation for the line in $\mathbb{A}^2$ parallel to $\mathbf{v}$ and passing through $S \cap T$ in each of the following cases:

- $\mathbf{v} = (2, 4)$, $S : 3x - 2y - 7 = 0$, $T : 2x + 3y = 0$,
- $\mathbf{v} = (-5\sqrt{2}, 7)$, $S : x - y = 0$, $T : x + y = 1$.

2.6. Let ABC be a triangle in the affine space $\mathbb{A}^n$. Consider the points C' and B' on the sides AB and AC respectively, such that

$$\overrightarrow{AC'} = \lambda \overrightarrow{BC'} \quad \text{and} \quad \overrightarrow{AB'} = \mu \overrightarrow{CB'}.$$

The lines BB' and CC' meet in the point M . For a fixed but arbitrary point $O \in \mathbb{A}^n$, show that

$$\overrightarrow{OM} = \frac{\overrightarrow{OA} - \lambda \overrightarrow{OB} - \mu \overrightarrow{OC}}{1 - \lambda - \mu}.$$

Deduce a formula for $\overrightarrow{OG}$ where G is the centroid of the triangle.

2.7. In $\mathbb{A}^n$, consider the angle BOB' and the points $A \in [OB]$, $A' \in [OB']$. Show that

$$\begin{aligned} \overrightarrow{OM} &= m \frac{1-n}{1-mn} \overrightarrow{OA} + n \frac{1-m}{1-mn} \overrightarrow{OA'} \\ \overrightarrow{ON} &= m \frac{n-1}{n-m} \overrightarrow{OA} + n \frac{m-1}{m-n} \overrightarrow{OA'} \end{aligned}$$

where $M = AB' \cap A'B$ and $N = AA' \cap BB'$ and where $\overrightarrow{OB} = m \overrightarrow{OA}$ and $\overrightarrow{OB'} = n \overrightarrow{OA'}$.

2.8. Show that the midpoints of the diagonals of a complete quadrilateral are collinear.

2.9. Determine parametric equations for the plane π in the following cases:

- a) π contains the point $M(1, 0, 2)$ and is parallel to the vectors $\mathbf{a}_1(3, -1, 1)$ and $\mathbf{a}_2(0, 3, 1)$,
- b) π contains the points $A(-2, 1, 1)$, $B(0, 2, 3)$ and $C(1, 0, -1)$,
- c) π contains the point $A(1, 2, 1)$ and is parallel to $\mathbf{i}$ and $\mathbf{j}$,
- d) π contains the point $M(1, 7, 1)$ and is parallel coordinate plane Oyz ,
- e) π contains the points $M_1(5, 3, 4)$ and $M_2(1, 0, 1)$, and is parallel to the vector $\mathbf{a}(1, 3, -3)$,
- f) π contains the point $A(1, 5, 7)$ and the coordinate axis Ox .

2.10. Determine Cartesian equations for the plane π in the following cases:

- a) $\pi : x = 2 + 3u - 4v, y = 4 - v, z = 2 + 3u$;
- b) $\pi : x = u + v, y = u - v, z = 5 + 6u - 4v$.

2.11. Determine parametric equations for the plane π in the following cases:

- a) $3x - 6y + z = 0$;
- b) $2x - y - z - 3 = 0$.

2.12. With the assumptions in Example 1.21, give parametric equations and Cartesian equations for the line AB and the plane ACD both in the coordinate system $\mathcal{K}$ and in the coordinate system $\mathcal{K}'$.

2.13. Show that the points $A(1, 0, -1)$, $B(0, 2, 3)$, $C(-2, 1, 1)$ and $D(4, 2, 3)$ are coplanar.

2.14. Determine the relative positions of the planes in the following cases

- a) $\pi_1 : x + 2y + 3z - 1 = 0, \pi_2 : x + 2y - 3z - 1 = 0$.
- b) $\pi_1 : x + 2y + 3z - 1 = 0, \pi_2 : 2x + y + 3z - 2 = 0, \pi_3 : x + 2y + 3z + 2 = 0$.

2.15. Show that the planes

$$\pi_1 : 3x + y + z - 1 = 0, \quad \pi_2 : 2x + y + 3z + 2 = 0, \quad \pi_3 : -x + 2y + z + 4 = 0$$

have a point in common.

2.16. Show that the pairwise intersection of the planes

$$\pi_1 : 3x + y + z - 5 = 0, \quad \pi_2 : 2x + y + 3z + 2 = 0, \quad \pi_3 : 5x + 2y + 4z + 1 = 0$$

are parallel lines.

2.17. Determine parametric equations for the line ℓ in the following cases:

- a) ℓ contains the point $M_0(2, 0, 3)$ and is parallel to the vector $\mathbf{a}(3, -2, -2)$,
 b) ℓ contains the point $A(1, 2, 3)$ and is parallel to the Oz -axis,
 c) ℓ contains the points $M_1(1, 2, 3)$ and $M_2(4, 4, 4)$.

2.18. Give Cartesian equations for the lines ℓ in the previous exercise.

2.19. Determine parametric equations for the line contained in the planes $x + y + 2z - 3 = 0$ and $x - y + z - 1 = 0$.

2.20. Consider the lines $\ell_1 : x = 1 + t, y = 1 + 2t, z = 3 + t, t \in \mathbb{R}$ and $\ell_2 : x = 3 + s, y = 2s, z = -2 + s, s \in \mathbb{R}$. Show that ℓ_1 and ℓ_2 are parallel and find the equation of the plane determined by the two lines.

2.21. Determine parametric equations of the line passing through $P(5, 0, -2)$ and parallel to the planes $\pi_1 : x - 4y + 2z = 0$ and $\pi_2 : 2x + 3y - z + 1 = 0$.

2.22. Determine an equation of the plane containing $P(2, 0, 3)$ and the line $\ell : x = -1 + t, y = t, z = -4 + 2t, t \in \mathbb{R}$.

2.23. For the points $A(2, 1, -1)$ and $B(-3, 0, 2)$, determine an equation of the bundle of planes passing through A and B .

2.24. Determine the relative positions of the lines $x = -3t, y = 2 + 3t, z = 1, t \in \mathbb{R}$ and $x = 1 + 5s, y = 1 + 13s, z = 1 + 10s, s \in \mathbb{R}$.

2.25. Determine the parameter m for which the line $x = -1 + 3t, y = 2 + mt, z = -3 - 2t$ doesn't intersect the plane $x + 3y + 3z - 2 = 0$.

2.26. Determine the values a and d for which the line $\frac{x-2}{3} = \frac{y+1}{2} = \frac{z-3}{-2}$ is contained in the plane $ax + y - 2z + d = 0$.

2.27. In each of the following, find a Cartesian equation of the plane in $\mathbb{A}^3$ passing through Q and parallel to the lines ℓ and ℓ' :

- a) $Q(1, -1, -2), \ell : x - y = 1, x + z = 5, \ell' : x = 1, z = 2$
 b) $Q(0, 1, 3), \ell : x + y = -5, x - y + 2z = 0, \ell' : 2x - 2y = 1, x - y + 2z = 1$

2.28. In each of the following, find the relative positions of the line ℓ and the plane π in $\mathbb{A}^3$, and, if they are incident, determine the point of intersection.

- a) $\ell : x = 1 + t, y = 2 - 2t, z = 1 - 4t, \pi : 2x - y + z - 1 = 0$
 b) $\ell : x = 2 - t, y = 1 + 2t, z = -1 + 3t, \pi : 2x + 2y - z + 1 = 0$

2.29. In each of the following, find a Cartesian equation for the plane in $\mathbb{A}^3$ containing the point Q and the line ℓ .

- a) $Q = (3, 3, 1), \ell : x = 2 + 3t, y = 5 + t, z = 1 + 7t$

b) $Q = (2, 1, 0)$, $\ell : x - y + 1 = 0, 3x + 5z - 7 = 0$

2.30. In each of the following, find Cartesian equations for the line ℓ in $\mathbb{A}^3$ passing through Q , contained in the plane π and intersecting the line ℓ'

a) $Q = (1, 1, 0)$, $\pi : 2x - y + z - 1 = 0$, $\ell' : x = 2 - t, y = 2 + t, z = t$

b) $Q = (-1, -1, -1)$, $\pi : x + y + z + 3 = 0$, $\ell' : x - 2z + 4 = 0, 2y - z = 0$

2.31. In each of the following, find Cartesian equations for the line ℓ in $\mathbb{A}^3$ passing through Q and coplanar to the lines ℓ' and ℓ'' . Furthermore, establish whether ℓ meets or is parallel to ℓ' and ℓ''

a) $Q = (1, 1, 2)$, $\ell' : 3x - 5y + z = -1, 2x - 3z = -9$, $\ell'' : x + 5y = 3, 2x + 2y - 7z = -7$

b) $Q = (2, 0, -2)$, $\ell' : -x + 3y = 2, x + y + z = -1$, $\ell'' : x = 2 - t, y = 3 + 5t, z = -t$

2.32. In each of the following, find the value of the real parameter k for which the lines ℓ and ℓ' are coplanar. Find a Cartesian equation for the plane that contains them, and find the point of intersection whenever they meet

a) $\ell : x = k + t, y = 1 + 2t, z = -1 + kt$, $\ell' : x = 2 - 2t, y = 3 + 3t, z = 1 - t$

b) $\ell : x = 3 - t, y = 1 + 2t, z = k + t$, $\ell' : x = 1 + t, y = 1 + 2t, z = 1 + 3t$

2.33. Find a Cartesian equation for the plane π in $\mathbb{A}^3$ which contains the line of intersection of the two planes

$$x + y = 3 \quad \text{and} \quad 2y + 3z = 4$$

and is parallel to the vector $\mathbf{v} = (3, -1, 2)$.

2.34. In the affine space $\mathbb{A}^4$ consider

$$\text{the plane } \alpha = \left\langle \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix} \right\rangle + \begin{bmatrix} 2 \\ 4 \\ 1 \\ 2 \end{bmatrix} \quad \text{and the line } \beta = \left\langle \begin{bmatrix} 1 \\ 1 \\ -1 \\ 1 \end{bmatrix} \right\rangle + \begin{bmatrix} 2 \\ 3 \\ -1 \\ 1 \end{bmatrix}.$$

Determine $\alpha \cap \beta$.

2.35. In $\mathbb{A}^4$ consider the affine subspaces

$$\alpha = \begin{bmatrix} 2 \\ 1 \\ 2 \\ 1 \end{bmatrix} \quad \beta = \left\langle \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right\rangle + \begin{bmatrix} 1 \\ 3 \\ 0 \\ 0 \end{bmatrix} \quad \gamma = \left\langle \begin{bmatrix} 2 \\ 1 \\ 3 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 2 \\ -2 \end{bmatrix} \right\rangle + \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} \quad \delta = \left\langle \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right\rangle.$$

Which of the following is true?

- | | | |
|------------------------|------------------------------|------------------------------|
| a) $\alpha \in \beta$ | d) $\beta \parallel \gamma$ | g) $\beta \subseteq \gamma$ |
| b) $\alpha \in \gamma$ | e) $\beta \parallel \delta$ | h) $\gamma \subseteq \delta$ |
| c) $\alpha \in \delta$ | f) $\gamma \parallel \delta$ | i) $\beta \subseteq \delta$ |

2.36. Consider the following affine subspaces of $\mathbb{A}^4$

$$Y : \begin{cases} x_1 + x_3 - 2 = 0 \\ 2x_1 - x_2 + x_3 + 3x_4 - 1 = 0 \end{cases}$$

$$Z : \begin{cases} x_1 + x_2 + 2x_3 - 3x_4 = 1 \\ x_2 + x_3 - 3x_4 = -1 \\ x_1 - x_2 + 3x_4 = 3 \end{cases}$$

- Determine the dimensions of Y and Z .
- Find parametric equations for each of the two affine subspaces.
- Is $Y \parallel Z$?

2.37. In Section 2.2.2 we deduce a linear equation for a plane in $\mathbb{A}^3$ via a determinant. What is the analogue of this description for lines? I.e. deduce Cartesian equations for lines starting from linear dependence of vectors (both in $\mathbb{A}^2$ and $\mathbb{A}^3$).

2.38. Consider the affine space $\mathbb{A}^3$. Show that if a line ℓ doesn't intersect a plane π then $\ell \parallel \pi$ in the sense of the Definition 2.14. Moreover, give an example in $\mathbb{A}^4$ of a line and a plane which do not intersect and which are not parallel.

2.39. Consider the affine space $\mathbb{A}^4$. Describe the relative positions of two planes.

2.40. In $\mathbb{A}^3$ discuss the relative positions of a plane and a line in terms of their Cartesian equations.

2.41. In $\mathbb{A}^3$ discuss the relative positions two lines in terms of their Cartesian equations.

$$\mathcal{K} = (0, \mathcal{B})$$

reference system for $\mathbb{E}^n$

$$0 \in \mathbb{E}^n$$

$\mathcal{B}$ basis of V^n

$$\forall \vec{p} \in \mathbb{E}^n, \boxed{[\vec{p}]_{\mathcal{K}} := [\vec{OP}]_{\mathcal{B}}}$$

Notations: V, V' - vector spaces

$\mathcal{B}, \mathcal{B}'$ - bases of V and V'

$f: V \rightarrow V'$ - linear map

$$M_{\mathcal{B}', \mathcal{B}}^{-1} = M_{\mathcal{B}, \mathcal{B}'}$$

$$M_{\mathcal{B}, \mathcal{B}}(f) = [f]_{\mathcal{B}, \mathcal{B}} = ([f(v_1)]_{\mathcal{B}} \dots [f(v_n)]_{\mathcal{B}}) \text{ where } \mathcal{B} = (v_1, \dots, v_n)$$

V - vector space

$\mathcal{B}, \mathcal{B}'$ bases of V

The base-change matrix λ from $\mathcal{B}$ to $\mathcal{B}'$ is:

$$M_{\mathcal{B}', \mathcal{B}} := M_{\mathcal{B}', \mathcal{B}}(\text{id}) = [\text{id}]_{\mathcal{B}', \mathcal{B}} = ([v_1]_{\mathcal{B}'} [v_2]_{\mathcal{B}'} \dots [v_n]_{\mathcal{B}'})$$

we use this in the following way, $\forall v \in V: \boxed{[v]_{\mathcal{B}'} = M_{\mathcal{B}', \mathcal{B}} \cdot [v]_{\mathcal{B}}}$

$$\text{ex: } M_{\mathcal{B}_1, \mathcal{B}_3}(f \circ \psi) = M_{\mathcal{B}_1, \mathcal{B}_2}(f) \cdot M_{\mathcal{B}_2, \mathcal{B}_3}(\psi)$$

$$\mathcal{K} = (0, \mathcal{B}), \mathcal{K}' = (0', \mathcal{B}')$$

$$\begin{aligned} [\vec{p}]_{\mathcal{K}'} &= [\vec{OP}]_{\mathcal{B}'} = M_{\mathcal{B}', \mathcal{B}} \cdot [\vec{OP}]_{\mathcal{B}} \\ &= M_{\mathcal{B}', \mathcal{B}} \cdot [\vec{OP} - \vec{OO'}]_{\mathcal{B}} \\ &= M_{\mathcal{B}', \mathcal{B}} \cdot ([\vec{OP}]_{\mathcal{B}} - [\vec{OO'}]_{\mathcal{B}}) \\ &= M_{\mathcal{B}', \mathcal{B}} \cdot ([\vec{p}]_{\mathcal{K}} - [\vec{O'}]_{\mathcal{K}}) \\ &= M_{\mathcal{B}', \mathcal{B}} \cdot [\vec{p}]_{\mathcal{K}} - M_{\mathcal{B}', \mathcal{B}} [\vec{OO'}]_{\mathcal{B}} \\ &= M_{\mathcal{B}', \mathcal{B}} [\vec{p}]_{\mathcal{K}} - [\vec{OO'}]_{\mathcal{B}'} \\ &= M_{\mathcal{B}', \mathcal{B}} [\vec{p}]_{\mathcal{K}} + [\vec{O'O}]_{\mathcal{B}'} \\ &= M_{\mathcal{B}', \mathcal{B}} [\vec{p}]_{\mathcal{K}} + [\vec{0}]_{\mathcal{K}'} \end{aligned}$$

From now on: $M_{\mathcal{B}', \mathcal{B}} =: M_{\mathcal{K}', \mathcal{K}}$

1.16

$$\mathcal{K} = (0, \vec{i}, \vec{j})$$

$$\mathcal{K}' = (0', \vec{i}', \vec{j}') \quad \text{reference systems for } \mathbb{E}^2$$

$$[0']_{\mathcal{K}} = \begin{pmatrix} 7 \\ -1 \end{pmatrix}$$

$$\begin{cases} \vec{i}' = -2\vec{i} + \vec{j} \\ \vec{j}' = \vec{i} + 2\vec{j} \end{cases} \Rightarrow \vec{i}' = \begin{bmatrix} -2 \\ 1 \end{bmatrix}_{\mathcal{K}}$$

$$\vec{j}' = \begin{bmatrix} 1 \\ 2 \end{bmatrix}_{\mathcal{K}}$$

$$[A]_{\mathcal{K}} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, [B]_{\mathcal{K}} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}, [C]_{\mathcal{K}} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

Find the base change matrix from $\mathcal{K}$ to $\mathcal{K}'$ and $[A]_{\mathcal{K}'}$, $[B]_{\mathcal{K}'}$, $[C]_{\mathcal{K}'}$.

Find the base change matrix from $\mathcal{K}'$ to $\mathcal{K}$ and check the previous result.

$$[A]_{\mathcal{K}'} = [\vec{OA}]_{\mathcal{K}'} = [\vec{OA}]_{\mathcal{K}} - [\vec{OO}']_{\mathcal{K}}$$

$$[\vec{OA}]_{\mathcal{K}'} = M_{\mathcal{K}', \mathcal{K}} [\vec{OA}]_{\mathcal{K}}$$

$$\begin{aligned} [\vec{OA}]_{\mathcal{K}'} &= M_{\mathcal{K}', \mathcal{K}} ([\vec{OA}]_{\mathcal{K}} - [\vec{OO}']_{\mathcal{K}}) = M_{\mathcal{K}', \mathcal{K}} [\vec{OA}]_{\mathcal{K}} - M_{\mathcal{K}', \mathcal{K}} [\vec{OO}']_{\mathcal{K}} \\ &= M_{\mathcal{K}', \mathcal{K}} [\vec{OA}]_{\mathcal{K}} - [\vec{OO}']_{\mathcal{K}'} \end{aligned}$$

$$[\vec{OA}]_{\mathcal{K}'} = M_{\mathcal{K}', \mathcal{K}} \begin{bmatrix} 1 \\ 2 \end{bmatrix} - [\vec{OO}']_{\mathcal{K}'}$$

$$[A]_{\mathcal{K}'} = M_{\mathcal{K}', \mathcal{K}} ([A]_{\mathcal{K}} - [0']_{\mathcal{K}}) = M_{\mathcal{K}', \mathcal{K}} [A]_{\mathcal{K}} + [0]_{\mathcal{K}'}$$

$$M_{\mathcal{K}, \mathcal{K}'} = \begin{bmatrix} -2 & 1 \\ 1 & 2 \end{bmatrix}^{-1} = \frac{1}{\Delta} \cdot A^* = \frac{1}{-5} \begin{bmatrix} 2 & -1 \\ -1 & -2 \end{bmatrix}$$

I did it

$$[A]_{\mathcal{K}'} = \begin{bmatrix} \frac{-2}{5} & \frac{1}{5} \\ \frac{1}{5} & \frac{2}{5} \end{bmatrix} \left(\begin{bmatrix} 1 \\ 2 \end{bmatrix} - \begin{bmatrix} 7 \\ -1 \end{bmatrix} \right) = \begin{bmatrix} \frac{-2}{5} & \frac{1}{5} \\ \frac{1}{5} & \frac{2}{5} \end{bmatrix} \begin{bmatrix} -6 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$$

$$\begin{aligned} [B]_{\mathcal{K}'} &= [\vec{OB}]_{\mathcal{K}'} = [\vec{OB}]_{\mathcal{K}'} - [\vec{OO}']_{\mathcal{K}'} = M_{\mathcal{K}', \mathcal{K}} ([\vec{OB}]_{\mathcal{K}} - [\vec{OO}']_{\mathcal{K}}) \\ &= M_{\mathcal{K}', \mathcal{K}} ([B]_{\mathcal{K}} - [0]_{\mathcal{K}}) \\ &= \begin{bmatrix} \frac{-2}{5} & \frac{1}{5} \\ \frac{1}{5} & \frac{2}{5} \end{bmatrix} \left(\begin{bmatrix} 2 \\ 4 \end{bmatrix} - \begin{bmatrix} 7 \\ -1 \end{bmatrix} \right) = \begin{bmatrix} \frac{-2}{5} & \frac{1}{5} \\ \frac{1}{5} & \frac{2}{5} \end{bmatrix} \begin{bmatrix} -5 \\ 5 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix} \end{aligned}$$

$$\begin{aligned}
 [C]_{K'} &= [\vec{O'C}]_{K'} = [\vec{OC}]_{K'} - [\vec{OO'}]_{K'} = M_{K'K} ([\vec{OC}]_K - [\vec{OO'}]_K) \\
 &= \begin{bmatrix} -2/5 & 1/5 \\ 1/5 & 2/5 \end{bmatrix} \left(\begin{bmatrix} 3 \\ 1 \end{bmatrix} - \begin{bmatrix} 7 \\ -1 \end{bmatrix} \right) = \begin{bmatrix} -2/5 & 1/5 \\ 1/5 & 2/5 \end{bmatrix} \begin{bmatrix} -4 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}
 \end{aligned}$$

Now the other way around, from K to K'

$$\begin{aligned}
 [A]_K &= [\vec{OA}]_K = M_{KK'} ([\vec{O'A}]_{K'} - [\vec{O'O}]_{K'}) = M_{KK'} \cdot [\vec{O'A}]_{K'} - M_{KK'} \cdot [\vec{O'O}]_{K'} \\
 &= M_{KK'} [A]_{K'} - M_{KK'} [O]_{K'}
 \end{aligned}$$

$$\begin{aligned}
 [O]_{K'} &= [\vec{O'O}]_{K'} = M_{K'K} [\vec{O'O}]_K = -M_{K'K} [\vec{OO'}]_K = -M_{K'K} [O']_K \\
 &= - \begin{bmatrix} -2/5 & 1/5 \\ 1/5 & 2/5 \end{bmatrix} \begin{bmatrix} 7 \\ -1 \end{bmatrix} = \begin{bmatrix} -14/5 & -1/5 \\ 7/5 & -2/5 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \end{bmatrix}
 \end{aligned}$$

$$\begin{bmatrix} -2 & 1 \\ 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 0 \end{bmatrix} - \begin{bmatrix} -2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} -6 \\ 3 \end{bmatrix} - \begin{bmatrix} -6-1 \\ 3-2 \end{bmatrix} = \begin{bmatrix} -6 \\ 3 \end{bmatrix} - \begin{bmatrix} -7 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

$$\begin{aligned}
 [B]_K &= [\vec{OB}]_K = [\vec{O'B}]_K - [\vec{O'O}]_K = M_{KK'} [\vec{O'B}]_{K'} + [\vec{OO'}]_K \\
 &= \begin{bmatrix} -2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} + \begin{bmatrix} 7 \\ -1 \end{bmatrix} = \begin{bmatrix} -5 \\ 5 \end{bmatrix} + \begin{bmatrix} 7 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}
 \end{aligned}$$

$$\begin{aligned}
 [C]_K &= [\vec{O'C}]_K - [\vec{O'O}]_K = M_{KK'} [\vec{O'C}]_{K'} + M_{KK'} [\vec{OO'}]_{K'} = M_{KK'} [\vec{OC}]_{K'} + [O']_K \\
 &= \begin{bmatrix} -2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 7 \\ -1 \end{bmatrix} = \begin{bmatrix} -4 \\ 2 \end{bmatrix} + \begin{bmatrix} 7 \\ -1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}
 \end{aligned}$$

1.17 ABCD tetrahedron

$$K_A = (A, \vec{AB}, \vec{AC}, \vec{AD})$$

$$K_B = (B, \vec{BA}, \vec{BC}, \vec{BD})$$

$$K'_A = (A, \vec{AB}, \vec{AD}, \vec{AC})$$

M midpoint of $[BC]$

(a) Write the coordinates of A, B, C, D, M in the 3 reference systems

(b) Write the base change matrix from K_A to K'_A

(c) Write the base change matrix from K_B to K_A

$$[A]_{K_A} = [\vec{AA}]_{K_A} = 0 \cdot [\vec{AB}]_{K_A} + 0 [\vec{AD}]_{K_A} + 0 [\vec{AC}]_{K_A} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}_A$$

$$[B]_{K_A} = [\vec{AB}]_{K_A} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}_{K_A}$$

$$[C]_{K_A} = [\vec{AC}]_{K_A} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}_{K_A}$$

$$[D]_{K_A} = [\vec{AD}]_{K_A} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}_{K_A}$$

$$[M]_{K_A} = [\vec{AM}]_{K_A} = \frac{1}{2} [\vec{AB}]_{K_A} + [\vec{AC}]_{K_A} = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1/2 \\ 1/2 \\ 0 \end{pmatrix}$$

$$[A]_{K'A} = [\vec{AA}]_{K'A} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$[B]_{K'A} = [\vec{AB}]_{K'A} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$[C]_{K'A} = [\vec{AC}]_{K'A} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\vec{r}_M = \frac{\vec{r}_B + \vec{r}_C}{2} = \frac{\vec{AB}}{2} + \frac{\vec{AC}}{2} \Rightarrow [M]_{K'A} = [\vec{AM}]_{K'A} = \begin{pmatrix} 1/2 \\ 0 \\ 1/2 \end{pmatrix}$$

$$[A]_{K_B} = [\vec{BA}]_{K_B} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$[B]_{K_B} = [\vec{BB}]_{K_B} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$[C]_{K_B} = [\vec{BC}]_{K_B} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$[D]_{K_B} = [\vec{BD}]_{K_B} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$[M]_{K_B} = [\vec{BM}]_{K_B} = \frac{1}{2} [\vec{BC}]_{K_B} = \begin{pmatrix} 0 \\ 1/2 \\ 0 \end{pmatrix}$$

$$b) M_{K'A, K_A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$c) M_{K_A K_B} = \begin{bmatrix} -1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

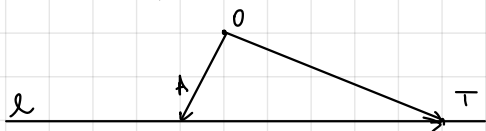
Seminar 3

Affine variety: $\mathcal{A} = a + U = \{a + \vec{v} \mid \vec{v} \in U\}$
 $a \in \mathbb{E}^n, U \leq \mathbb{R}^n$
 $\dim \mathcal{A} := \dim_{\mathbb{R}} U$

$U =: \mathcal{D}(\mathcal{A}) =$ direction
subspace of $\mathcal{A}$

If $\dim \mathcal{A} = 1 \Rightarrow \mathcal{A}$ linear
 $\dim \mathcal{A} = 2 \Rightarrow \mathcal{A}$ plane

l line, $A \in l, \vec{v} \in \mathcal{D}(l)$ (ie. $\vec{v} \parallel l$)



Fix an origin O

$$\vec{r}_T = \vec{r}_A + \vec{AT}$$

$$\vec{AT} \in \mathcal{D}(l) \Rightarrow \exists \lambda \in \mathbb{R}: \vec{AT} = \lambda \vec{v}$$

$$\vec{r}_T = \vec{r}_A + \lambda \vec{v}, \lambda \in \mathbb{R}$$

(vector eq of l)

Fix a reference system $\mathcal{K} = (O, \mathcal{B})$, suppose $n=2$

$$[T]_{\mathcal{K}} = \begin{pmatrix} x \\ y \end{pmatrix}, [A]_{\mathcal{K}} = \begin{pmatrix} x_A \\ y_A \end{pmatrix}$$

$$[\vec{v}]_{\mathcal{K}} = \begin{pmatrix} x_{\vec{v}} \\ y_{\vec{v}} \end{pmatrix}$$

$$l: \begin{cases} x = x_A + \lambda x_{\vec{v}} \\ y = y_A + \lambda y_{\vec{v}} \end{cases} \quad \text{parametric eq. of } l$$

$$\text{If } x_{\vec{v}}, y_{\vec{v}} \neq 0 \Rightarrow \frac{x - x_A}{x_{\vec{v}}} = \frac{y - y_A}{y_{\vec{v}}}$$

$$\text{If } x_{\vec{v}} = 0 \Rightarrow l: x = x_A$$

$$\text{If } y_{\vec{v}} = 0 \Rightarrow l: y = y_A$$

- implicit form: $y_{\vec{v}}(x - x_A) - x_{\vec{v}}(y - y_A) = 0$
 $Ax + By + C = 0$

- explicite form:

$$\text{If } AB \neq 0 \Rightarrow y = -\frac{A}{B}x - \frac{C}{B}$$

$$\text{If } A=0, B \neq 0 \Rightarrow y = -\frac{C}{B}$$

$$\text{If } A \neq 0, B=0 \Rightarrow x = -\frac{C}{A}$$

$$\mathcal{D}(l) = \langle \vec{n} \rangle = \langle v_1, v_2, \dots, v_n \rangle = \{ \alpha_1 v_1 + \dots + \alpha_n v_n \mid \alpha \in K \}$$
$$\langle v \rangle = \{ \alpha v \mid \alpha \in K \}$$

2.1 / 2.2 Determine the parametric and cartesian eq. for the line l and its direction subspace.

a) $l \ni A(1, 2)$, $l \parallel \vec{a}(3, -1)$

b) $l \ni O(0, 0)$, $l \parallel \vec{b}(4, 5)$

c) $l \ni M(1, 7)$, $l \parallel O_y$

d) $l \ni M(2, 4)$, $M(2, -5)$

a) parametric eq. $l: \begin{cases} x = x_A + \lambda x_{\vec{n}} \\ y = y_A + \lambda y_{\vec{n}} \end{cases}$

$$l: \begin{cases} x = 1 + \lambda 3 \\ y = 2 - \lambda \end{cases}$$

$$\Rightarrow \frac{x-1}{3} = 2-y \Rightarrow x-1 = 6-3y \Rightarrow x+3y-7=0 \text{ (implicit)}$$

$$y = -\frac{x}{3} + \frac{7}{3} \text{ (explicit)}$$

$$\mathcal{D}(l) = \langle \vec{a} \rangle = \langle (3, -1) \rangle$$

b) $l \ni O(0, 0)$, $l \parallel \vec{b}(4, 5)$

$$l: \begin{cases} x = 4\lambda \\ y = 5\lambda \end{cases} \Rightarrow \frac{x}{4} = \frac{y}{5} \Rightarrow 5x - 4y = 0$$
$$y = \frac{5}{4}x$$

$$\mathcal{D}(l) = \langle \vec{b} \rangle = \langle (4, 5) \rangle$$

c) $l \ni M(1, 7)$, $l \parallel O_y$

$$l: \begin{cases} x = 1 \\ y = 7 + 2 \end{cases}$$

$$l: x = 1$$

$$\mathcal{D}(l) = \langle (0, 1) \rangle$$

$$d) M(2, 4), N(2, -5) \in l$$

$$\vec{MN} = (2-2)\vec{i} + (-5-4)\vec{j} = 0 - 9\vec{j}$$

$$\vec{MN}(0, -9)$$

$$l: \begin{cases} x = 2 \\ y = 4 - 9x \end{cases}$$

$$l: x = 2$$

$$\mathcal{D}(l) = \langle (0, -9) \rangle$$

2.5 Determine the eq. of the line l parallel to $\vec{n}$ and passing through SNT if:

$$\vec{n} = (2, 4)$$

$$S: 3x - 2y - 7 = 0 \quad + \text{ write } \mathcal{D}(l)$$

$$T: 2x + 3y = 0$$

$$\begin{cases} 3x - 2y - 7 = 0 \quad | \cdot 3 \\ 2x + 3y = 0 \quad | \cdot 2 \end{cases} \Rightarrow \begin{cases} 9x - 6y - 21 = 0 \\ 4x + 6y = 0 \end{cases} (+)$$

$$13x = 21 \Rightarrow x = \frac{21}{13}$$

$$\frac{42}{13} + 3y = 0 \Rightarrow y = -\frac{42}{39} = -\frac{14}{13}$$

$$\left. \begin{array}{l} 13x = 21 \Rightarrow x = \frac{21}{13} \\ \frac{42}{13} + 3y = 0 \Rightarrow y = -\frac{14}{13} \end{array} \right\} \xrightarrow{\text{we denote}} A\left(\frac{21}{13}, -\frac{14}{13}\right)$$

$$l: \begin{cases} x = \frac{21}{13} + 2\lambda \\ y = -\frac{14}{13} + 4\lambda \end{cases}$$

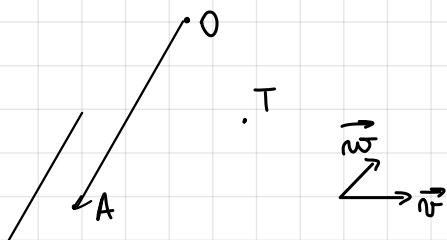
$$\Rightarrow \frac{x - \frac{21}{13}}{2} = \frac{y + \frac{14}{13}}{4} \Rightarrow 4x - \frac{84}{13} = 2y + \frac{28}{13}$$

$$\Rightarrow 4x - 2y - \frac{112}{13} = 0$$

$$\Rightarrow \mathcal{D}(l) = \langle \vec{n} \rangle = \langle (2, 4) \rangle = \langle (1, 2) \rangle$$

Π plane, $A \in \Pi$, $\vec{v}, \vec{w} \in \mathcal{D}(\Pi)$

linear indep.



$$\vec{r}_T = \vec{r}_A + \vec{AT}$$

$$\vec{AT} \in \mathcal{D}(\Pi) \Rightarrow \exists \lambda, \mu \in \mathbb{R}, \vec{AT} = \lambda \vec{v} + \mu \vec{w}$$

$$\Pi: \vec{r}_T = \vec{r}_A + \lambda \vec{v} + \mu \vec{w} \quad \leftarrow \text{vector equation}$$

Fix $\mathcal{K} = (O, \mathcal{B})$ a reference system. So:

$$\Pi: \begin{cases} x = x_A + \lambda x_{\vec{v}} + \mu x_{\vec{w}} \\ y = y_A + \lambda y_{\vec{v}} + \mu y_{\vec{w}} \\ z = z_A + \lambda z_{\vec{v}} + \mu z_{\vec{w}} \end{cases}$$

parametric equation

$$\Leftrightarrow \Pi: \begin{vmatrix} x-x_A & y-y_A & z-z_A \\ x_{\vec{v}} & y_{\vec{v}} & z_{\vec{v}} \\ x_{\vec{w}} & y_{\vec{w}} & z_{\vec{w}} \end{vmatrix} = 0 \quad (\text{symmetric form})$$

- implicate form: $Ax + By + Cz + D = 0$

- explicite form: if $c \neq 0$: $z = -\frac{A}{C}x - \frac{B}{C}y - \frac{D}{C}$

2.10 Determine Cartesian eq. for the plane Π :

$$a) \quad \Pi: \begin{cases} x = 2 + 3\mu - 4\nu \\ y = 4 - \nu \\ z = 2 + 3\mu \end{cases}$$

$$b) \quad \Pi: \begin{cases} x = \mu + \nu \\ y = \mu - \nu \\ z = 5 + 6\mu - 4\nu \end{cases}$$

$$a) \quad \begin{vmatrix} x-2 & y-4 & z-2 \\ 3 & 0 & 3 \\ -4 & -1 & 0 \end{vmatrix} = 0 - 3(z-2) - 12(y-4) - 0 - 0 + 3(x-2) \\ = -3z + 6 - 12y + 48 + 3x - 6 = 3x - 12y - 3z + 48$$

$$\Rightarrow x - 4y - z + 16 = 0$$

$$\mathcal{D}(\Pi) = \langle (3, 0, 3), (-4, -1, 0) \rangle$$

$$b) \begin{vmatrix} x & y & z-5 \\ 1 & 1 & 6 \\ 1 & -1 & -4 \end{vmatrix} = \begin{vmatrix} x & x+y & z-5+4x \\ 1 & 2 & 10 \\ \boxed{1} & 0 & 0 \end{vmatrix} = 5x + 5y - z + 5 - 4x = x + 5y - z + 5 = 0$$

$$\mathcal{D}(\Pi) = \langle (1, 1, 6), (1, -1, -4) \rangle$$

2.11 Determine parametric eq. for Π . a) $3x - 6y + z = 0$

$$b) 2x - y - z - 3 = 0$$

$$a) \begin{cases} x = 2y - \frac{1}{3}z = 2\lambda - \frac{1}{3}\mu \\ y = \lambda \\ z = \mu \end{cases}$$

$$\mathcal{D}(\Pi) = \langle (2, 1, 0), (-\frac{1}{3}, 0, 1) \rangle = \langle (2, 1, 0), (1, 0, -3) \rangle$$

$$b) \begin{cases} x = \frac{y}{2} + \frac{z}{2} + 3 = \frac{3}{2} + \alpha \frac{1}{2} + \beta \frac{1}{2} \\ y = \alpha \\ z = \beta \end{cases}$$

$$\mathcal{D}(\Pi) = \langle (\frac{1}{2}, 1, 0), (\frac{1}{2}, 0, 1) \rangle = \langle (1, 2, 0), (1, 0, 2) \rangle$$

Seminar 4

Dot product: (scalar product)

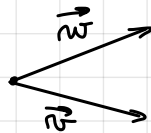
$$\vec{v}, \vec{w} \in V, \quad \vec{v} \cdot \vec{w} = \begin{cases} 0, & \text{if } \vec{v} = \vec{0} \text{ or } \vec{w} = \vec{0} \\ |\vec{v}| \cdot |\vec{w}| \cdot \cos(\angle(\vec{v}, \vec{w})) & \text{otherwise} \end{cases}$$

properties:

• bilinearity: $\forall \vec{v}_1, \vec{v}_2, \vec{w} \in V$

$$(\alpha \vec{v}_1 + \beta \vec{v}_2) \cdot \vec{w} = \alpha \vec{v}_1 \cdot \vec{w} + \beta \vec{v}_2 \cdot \vec{w}$$

- symmetry: $\forall \vec{u}, \vec{v} \in V$
 $\vec{u} \cdot \vec{v} = \vec{v} \cdot \vec{u}$



- positive definiteness: $\forall \vec{u} \in V$
 $\vec{u} \cdot \vec{u} \in \mathbb{R} \geq 0$ if $\vec{u} \neq \vec{0}$
then $\vec{u} \cdot \vec{u} > 0$

consequence:

- $\vec{u} \cdot \vec{u} = |\vec{u}|^2$
- $\vec{u}^2 - \vec{v}^2 = (\vec{u} - \vec{v}) \cdot (\vec{u} + \vec{v})$

- if we have an orthonormal basis $B = (\vec{u}_1, \dots, \vec{u}_n)$

$$\forall i \neq j: \vec{u}_i \cdot \vec{u}_j = 0$$

$$\forall i: |\vec{u}_i| = 1$$

$$\text{Let } \vec{w}, \vec{w}' \in V, [\vec{w}]_B = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, [\vec{w}']_B = \begin{pmatrix} x'_1 \\ x'_2 \\ \vdots \\ x'_n \end{pmatrix}$$

$$\vec{w} \cdot \vec{w}' = x_1 x'_1 + x_2 x'_2 + \dots + x_n x'_n$$

$$|\vec{w}| = \sqrt{x_1^2 + \dots + x_n^2}$$

2.1. $\vec{m}, \vec{n}$ unit vectors at $\angle(\vec{m}, \vec{n}) = 60^\circ$

Determine the length of the diagonals in the parallelogram spanned by

$$\vec{a} = 2\vec{m} + \vec{n}$$

$$\vec{b} = \vec{m} - 2\vec{n}$$

$$\vec{d}_1 = \vec{a} + \vec{b}$$

$$\begin{aligned} |\vec{a} + \vec{b}|^2 &= (\vec{a} + \vec{b}) \cdot (\vec{a} + \vec{b}) \\ &= (2\vec{m} + \vec{n} + \vec{m} - 2\vec{n}) \cdot (2\vec{m} + \vec{n} + \vec{m} - 2\vec{n}) \\ &= (3\vec{m} - \vec{n}) \cdot (3\vec{m} - \vec{n}) \\ &= 9\vec{m}^2 + \vec{n}^2 - 6\vec{m} \cdot \vec{n} \\ \vec{m}^2 &= |\vec{m}|^2 = 1 \\ \vec{n}^2 &= |\vec{n}|^2 = 1 \end{aligned}$$

$$\vec{m} \cdot \vec{n} = |\vec{m}| |\vec{n}| \cos(\angle(\vec{m}, \vec{n})) = 1 \cdot 1 \cdot \frac{1}{2}$$

$$\Rightarrow |d_1|^2 = 9 - 6 \cdot \frac{1}{2} + 1 = 7 \Rightarrow d_1 = \sqrt{7}$$

$$\vec{d}_2 = \vec{a} - \vec{b}$$

$$\begin{aligned} |\vec{a} - \vec{b}|^2 &= (\vec{a} - \vec{b}) \cdot (\vec{a} - \vec{b}) = (2\vec{m} + \vec{n} - \vec{m} + 2\vec{n}) \cdot (2\vec{m} + \vec{n} - \vec{m} + 2\vec{n}) \\ &= (\vec{m} + 3\vec{n}) \cdot (\vec{m} + 3\vec{n}) = \vec{m}^2 + 9\vec{n}^2 + 6\vec{m} \cdot \vec{n} \end{aligned}$$

$$\vec{m}\vec{n} = \frac{1}{2}$$

$$|d_2|^2 = 1 + 9 + 6 \cdot \frac{1}{2} = 13 \Rightarrow d_2 = \sqrt{13}$$

3.4. Let $(\vec{i}, \vec{j}, \vec{k})$ be an orthonormal basis. Consider the vectors:

$$\vec{q} = 3\vec{i} + \vec{j}$$

$$\vec{p} = \vec{i} + 2\vec{j} + 2\vec{k}$$

Determine λ such that $\cos(\angle(\vec{p}, \vec{q})) = \frac{5}{12}$

$$\vec{p}\vec{q} = 3 + 2 = 5$$

$$\underbrace{\vec{p}\vec{q}}_5 = |\vec{p}| \cdot |\vec{q}| \cdot \underbrace{\cos(\vec{p}, \vec{q})}_{\frac{5}{12}}$$

$$\Rightarrow 5 = |\vec{p}| \cdot |\vec{q}| \cdot \frac{5}{12}$$

$$\Rightarrow |\vec{p}| \cdot |\vec{q}| = 12$$

$$|\vec{q}| = \sqrt{9+1} = \sqrt{10}$$

$$|\vec{p}| = \sqrt{1+4+\lambda^2} = \sqrt{5+\lambda^2}$$

$$|\vec{p}| = \frac{12}{\sqrt{10}}$$

$$\sqrt{5+\lambda^2} = \frac{12}{\sqrt{10}} \Rightarrow 5+\lambda^2 = \frac{144}{10} \Rightarrow \lambda^2 = \frac{94}{10}$$

$$\lambda = \pm \sqrt{\frac{94}{10}}$$

3.6 $\triangle ABC$

$$\vec{AB}^2 + \vec{AC}^2 - \vec{BC}^2 = 2\vec{AB} \cdot \vec{AC} \quad (\text{cosine law})$$

proof: $\vec{AB}^2 - \vec{BC}^2 + \vec{AC}^2 = (\vec{AB} - \vec{BC})(\underbrace{\vec{AB} + \vec{BC}}_{\vec{AC}}) + \vec{AC}^2 = (\vec{AB} - \vec{BC} + \vec{AC}) \cdot \vec{AC}$

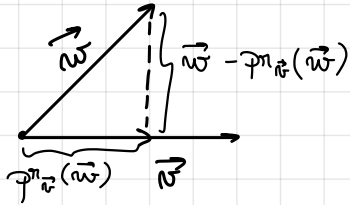
$$= (\vec{r}_B - \vec{r}_A - \vec{r}_C + \vec{r}_B + \vec{r}_C - \vec{r}_A) \cdot \vec{AC} = 2(\vec{r}_B - \vec{r}_A) \cdot \vec{AC} = 2\vec{AB} \cdot \vec{AC}$$

3.7. ABCD tetrahedron

$$\cos(\angle(\vec{AB}, \vec{CD})) = \frac{AB^2 + BC^2 - AC^2 - BD^2}{2AB \cdot CD} \quad (\text{cos. law in 3D})$$

$$2 \vec{AB} \cdot \vec{CD} = AD^2 - AC^2 + BC^2 - BD^2$$

$$\begin{aligned} (\vec{AD} - \vec{AC})(\vec{AD} + \vec{AC}) + (\vec{BC} - \vec{BD})(\vec{BC} + \vec{BD}) &= \vec{CD}(\vec{AD} + \vec{AC}) - \vec{CD}(\vec{BC} + \vec{BD}) \\ &= \vec{CD}(\vec{AD} - \vec{BD} + \vec{AC} - \vec{BC}) = \vec{CD}(\vec{AB} + \vec{AB}) = 2 \vec{AB} \cdot \vec{CD} \end{aligned}$$



$$|\text{pr}_{\vec{v}}(\vec{w})| = |\vec{w}| \cdot \cos(\angle \vec{v}, \vec{w})$$

$$\text{pr}_{\vec{v}}(\vec{w}) = \vec{v} \cdot \frac{|\text{pr}_{\vec{v}}(\vec{w})|}{|\vec{v}|} = \frac{|\vec{w}| \cdot \cos(\angle \vec{v}, \vec{w})}{|\vec{v}|} \cdot \vec{v} = \frac{\vec{v} \cdot \vec{w}}{\vec{v} \cdot \vec{v}} \cdot \vec{v}$$

Gram-Schmidt:

$B = (\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n)$ basis for V

I Orthogonalization

$$1. \vec{v}_1' = \vec{v}_1$$

$$2. \vec{v}_2' = \vec{v}_2 - \text{pr}_{\vec{v}_1'}(\vec{v}_2)$$

$$\begin{aligned} 3. \vec{v}_3' &= \vec{v}_3 - \text{pr}_{\text{span}(\vec{v}_1', \vec{v}_2')}(\vec{v}_3) = \vec{v}_3 - \text{pr}_{\vec{v}_1'}(\vec{v}_3) - \text{pr}_{\vec{v}_2'}(\vec{v}_3) \\ &= \vec{v}_3 - \frac{\vec{v}_3 \cdot \vec{v}_1'}{\vec{v}_1' \cdot \vec{v}_1'} \cdot \vec{v}_1' - \frac{\vec{v}_3 - \vec{v}_1'}{\vec{v}_2' \cdot \vec{v}_2'} \cdot \vec{v}_2' \end{aligned}$$

Repeat this until we get the orthogonal basis $B' = (\vec{v}_1', \vec{v}_2', \dots, \vec{v}_n')$

II Normalization

$$\vec{v}_{i1}' = \frac{\vec{v}_1'}{|\vec{v}_1'|} \Rightarrow B'' \text{ - orthonormal basis.}$$

3.10 In an orthonormal basis consider the vectors:

$$\vec{v}_1(0, 1, 0), \vec{v}_2(2, 1, 0), \vec{v}_3(-1, 0, 1)$$

Use the Gram-Schmidt process to find an orthonormal basis containing $\vec{v}_1$.

I
orthogonalization

$$\vec{v}_1' = \vec{v}_1 = (0, 1, 0)$$

$$\begin{aligned}\vec{v}_2' &= \vec{v}_2 - \text{proj}_{\vec{v}_1'}(\vec{v}_2) \\ &= \vec{v}_2 - \frac{\vec{v}_2 \cdot \vec{v}_1'}{\vec{v}_1' \cdot \vec{v}_1'} \cdot \vec{v}_1'\end{aligned}$$

$$= (2, 1, 0) - \frac{1}{1} \cdot (0, 1, 0)$$

$$= (2, 0, 0)$$

$$\vec{v}_3' = \vec{v}_3 - \frac{\vec{v}_3 \cdot \vec{v}_1'}{\vec{v}_1' \cdot \vec{v}_1'} \cdot \vec{v}_1' - \frac{\vec{v}_3 \cdot \vec{v}_2'}{\vec{v}_2' \cdot \vec{v}_2'} \cdot \vec{v}_2'$$

$$= (-1, 0, 1) - \frac{0}{1} \cdot (0, 1, 0) - \frac{(1, 0, 1) \cdot (2, 0, 0)}{4} \cdot (2, 0, 0)$$

$$= (-1, 0, 1) - 0 - \frac{-2}{4} \cdot (2, 0, 0)$$

$$= (-1, 0, 1) + (1, 0, 0) = (0, 0, 1)$$

II
normalization

$$\vec{v}_1'' = \frac{\vec{v}_1'}{|\vec{v}_1'|} = \frac{(0, 1, 0)}{1} = (0, 1, 0)$$

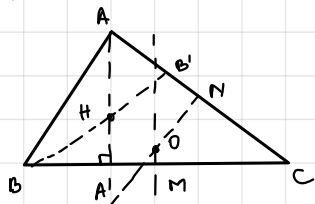
$$\vec{v}_2'' = \frac{\vec{v}_2'}{|\vec{v}_2'|} = \frac{(2, 0, 0)}{\sqrt{4}} = (1, 0, 0)$$

$$\vec{v}_3'' = \vec{v}_3' = (0, 0, 1)$$

3.18. Determine the circumcenter and the orthocenter of $\triangle ABC$
 $A(1, 2), B(3, -2), C(5, 6)$ (orthonormal ref. system)

circumcenter = $\cap$ (perpendicular bisectors $\leftrightarrow$ mediators)
 (centrul ceroului circumscris)

orthocenter = $\cap$ (heights)
 (centrul ceroului inscris)



$$m_{AB} = \frac{-2-2}{3-1} = -2$$

$$AA' \perp BC \Rightarrow m_{AA'} \cdot m_{BC} = -1$$

$$m_{BC} = \frac{6+2}{5-3} = 4 \Rightarrow m_{AA'} = -\frac{1}{4}$$

$$AA': y - y_A = m_{AA'} (x - x_A) \Rightarrow AA': y - y_A = m_{AA'} (x - x_A) \Rightarrow AA': y - 2 = -\frac{1}{4}(x - 1)$$

$$\Rightarrow AA': x + 4y - 9 = 0$$

$$m_{CC'} = \frac{1}{2}$$

$$CC': y - 6 = \frac{1}{2}(x - 5) \Rightarrow CC': x - 2y + 7 = 0$$

$$H: \begin{cases} x + 4y - 9 = 0 \Rightarrow x = 9 - 4y \\ x - 2y + 7 = 0 \end{cases} \Leftrightarrow \begin{cases} x = 9 - 4y \\ 16 = 6y \Rightarrow y = \frac{8}{3} \end{cases} \Rightarrow x = -\frac{5}{3}$$

$$\begin{aligned} m_{NO} &= m_{BB'} & m_{BB'} \cdot m_{AC} &= -1 \\ m_{MO} &= m_{AA'} & m_{AC} &= \frac{6-2}{5-1} = 1 \end{aligned} \Rightarrow m_{BB'} = -1$$

$$NO: y - y_N = m_{NO}(x - x_N) \quad , \quad x_N = \frac{x_A + x_C}{2} = 4$$

$$NO: y - 3 = -(x - 4) \quad y_N = \frac{y_A + y_C}{2} = 3$$

$$NO: x + y - 7 = 0$$

$$MO: y - y_M = m_{MO}(x - x_M) \quad , \quad x_M = 4$$

$$MO: y - 2 = -\frac{1}{4}(x - 4) \quad y_M = 2$$

$$MO: x + 4y - 12 = 0$$

$$O: \begin{cases} x + y - 7 = 0 \\ x + 4y - 12 = 0 \end{cases} \Rightarrow \begin{aligned} x &= \frac{16}{3} \\ y &= \frac{5}{3} \end{aligned}$$

Seminar 5 · 26, 29, 30, 31, 32, 33, 35, 36, 38, 39, 42

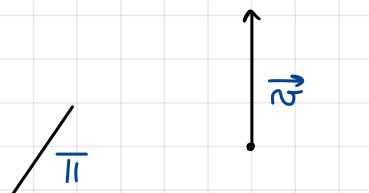
We will work with respect to the orthonormal reference system.

$$L: ax + by + c = 0 \text{ (in } \mathbb{3D}\text{)}$$

$\vec{n}(a, b)$ normal vector for L

$$\Pi: Ax + By + C \in \mathbb{3D}$$

$\Rightarrow \vec{n}(A, B, C)$ normal vector for Π

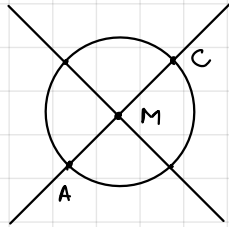
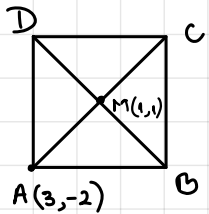


$$\Pi: Ax + By + Cz + D = 0$$

$$P(x_0, y_0, z_0)$$

$$\text{dist}(P, \Pi) = \frac{|Ax_0 + By_0 + Cz_0 + D|}{\sqrt{A^2 + B^2 + C^2}}$$

3.26. The point $A(3, -2)$ is the vertex of a square $ABCD$ and $M(1, 1)$ is the intersection point of the diagonals. Determine cartesian equations for the sides of the square.



$$M\left(\frac{x_A + x_C}{2}, \frac{y_A + y_C}{2}\right) = M(1, 1)$$

$$1 = \frac{3 + x_C}{2} \Rightarrow x_C = -1$$

$$1 = \frac{-2 + y_C}{2} \Rightarrow y_C = 4 \quad \Rightarrow C(-1, 4)$$

$$A(3, -2), C(-1, 4)$$

$$\begin{aligned} AC: mx + n &= y \Leftrightarrow \begin{aligned} 3m + n &= -2 \\ -m + n &= 4 \end{aligned} \\ \hline 4m &= -6 \Rightarrow m = -\frac{3}{2} \end{aligned}$$

$$n = 4 - \frac{3}{2} = \frac{5}{2}$$

$$\begin{aligned} \Rightarrow AC: -\frac{3}{2}x + \frac{5}{2} &= y \\ -3x + 5 &= 2y \end{aligned}$$

$$AC: -3x + 5 = 2y$$

$$\begin{aligned} MB \perp AC &\Rightarrow m_{AC} \cdot m_{BM} = -1 \Rightarrow m_{BM} = \frac{2}{3} \\ &\downarrow \\ &-\frac{3}{2} \end{aligned}$$

$$\begin{aligned} BM: mx + n &= y, \quad M(1, 1) \\ \frac{2}{3}x + n &= y \end{aligned}$$

$$2x + 3n = 3y$$

$$2 + 3n = 3 \Rightarrow 3n = 1 \Rightarrow n = \frac{1}{3}$$

$$BM: 2x + n = 3y$$

The eq. of the circle centered is $\Omega(x_0, y_0)$ with radius R is: $(x-x_0)^2 + (y-y_0)^2 = R^2$

$C(-1, 4)$

$AM = \sqrt{(1-3)^2 + (1+2)^2} = \sqrt{4+9} = \sqrt{13}$

$m_{AC} = -\frac{3}{2} \Rightarrow m_{BD} = \frac{2}{3}$

$BD: y - y_M = m_{BD}(x - x_M)$

$y - 1 = \frac{2}{3}(x - 1) \Leftrightarrow 2x + 1 = 3y$

$BD: \begin{cases} 2x - 3y + 1 = 0 \\ (x-1)^2 + (y-1)^2 = 13 \end{cases}$

$BD: \begin{cases} x = \frac{3y-1}{2} \\ \left(\frac{3y-1}{2} - 1\right)^2 + (y-1)^2 = 13 \end{cases}$

$\left(\frac{3y-3}{2}\right)^2 + (y-1)^2 = 13$

$\dots (y-1)^2 \cdot 13 = 52 \Rightarrow (y-1)^2 = 4 \Rightarrow y = \{3, -1\}$

For $y = 3 \Rightarrow 2x - 8 = 0 \Rightarrow x = 4$

For $y = -1 \Rightarrow 2x + 4 = 0 \Rightarrow x = -2$

So, $B(4, 3), D(-2, -1)$ or $B(-2, -1), D(4, 3)$

$AB: \frac{x - x_A}{x - x_B} = \frac{y - y_A}{y - y_B} \dots$

$AB: y = 5x - 17$

$m_{AB} = 5 \Rightarrow CD: y - y_C = 5(x - x_C)$

$CD: y = 5x + 9$

$m_{BC} = -\frac{1}{5}$

$BC: y - y_B = -\frac{1}{5}(x - x_B)$

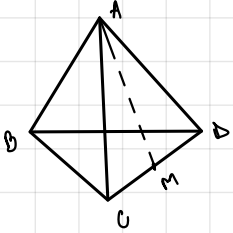
$BC: x + 5y - 19 = 0$

$m_{AD} = -\frac{1}{5}$

$$AD: y+1 = -\frac{1}{5}(x+2)$$

$$AD: 5y + x + 7 = 0$$

3.30 $A(2,1,0)$, $B(1,3,5)$, $C(6,3,4)$, $D(0,-7,8)$. Vertex of a tetrahedron. Determine a cartesian equation for the plane π that contains $[AB]$ and the midpoint M of $[CD]$.



$$x_M = \frac{x_C + x_D}{2}; \quad x_M = 3$$

$$y_M = \frac{y_C + y_A}{2}; \quad y_M = -2$$

$$z_M = \frac{z_C + z_A}{2}; \quad z_M = 6$$

$$\vec{BM} = (x_M - x_B)\vec{i} + (y_B - y_M)\vec{j} + (z_M - z_B)\vec{k}$$

$$\vec{BM} = 2\vec{i} - 5\vec{j} + \vec{k}$$

$$\vec{AB} = (x_B - x_A)\vec{i} + (y_B - y_A)\vec{j} + (z_B - z_A)\vec{k}$$

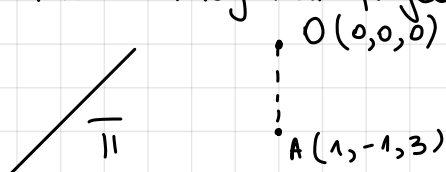
$$\vec{AB} = -\vec{i} + 2\vec{j} + 5\vec{k}$$

$$\pi: \begin{vmatrix} x - x_M & y - y_M & z - z_M \\ x_{\vec{BM}} & y_{\vec{BM}} & z_{\vec{BM}} \\ x_{\vec{AB}} & y_{\vec{AB}} & z_{\vec{AB}} \end{vmatrix} = 0$$

$$\pi: \begin{vmatrix} x-3 & y+2 & z-6 \\ 2 & -5 & 1 \\ -1 & 2 & 5 \end{vmatrix} = 0$$

$$= (\text{regula } \Delta - \text{loi}) - 27x - 11y - z + 65 = 0$$

3.32 Determine a Cartesian equation for the plane π if $A(1,-1,3)$ is the orthogonal projection of the origin on π .



$$\vec{OA}(1, -1, 3)$$

$$\overline{\Pi} : x - y + 3z + d = 0$$

$$A \in \overline{\Pi}$$

$$1 + 1 + 9 + d = 0 \Rightarrow d = -11$$

$$\Rightarrow \overline{\Pi} : x - y + 3z - 11 = 0$$

3.33 Determine the distance between the planes :

$$\overline{\Pi}_1 : x - 2y - 2z + 7 = 0$$

$$\overline{\Pi}_2 : 2x - 4y - 4z + 17 = 0$$

$$\begin{aligned} \vec{n}_{\overline{\Pi}_1} & (1, -2, -2) \\ \vec{n}_{\overline{\Pi}_2} & (2, -4, -4) = 2\vec{n}_{\overline{\Pi}_1} \end{aligned} \quad \left\{ \Rightarrow \overline{\Pi}_1 \parallel \overline{\Pi}_2 \right.$$

$$d(\overline{\Pi}_1, \overline{\Pi}_2) = d(P, \overline{\Pi}_2), \forall P \in \overline{\Pi}_1$$

$$P(-7, 0, 0)$$

$$d(P, \overline{\Pi}_2) = \frac{|-14 + 17|}{\sqrt{36}} = \frac{3}{6} = \frac{1}{2}$$

3.37 Determine the values a and c for which the line l :

$$\begin{cases} 3x - 2y + z + 3 = 0 \\ 4x - 3y + 4z + 1 = 0 \end{cases}$$

is perpendicular to the plane $\overline{\Pi} : ax + 8y + cz + 2 = 0$

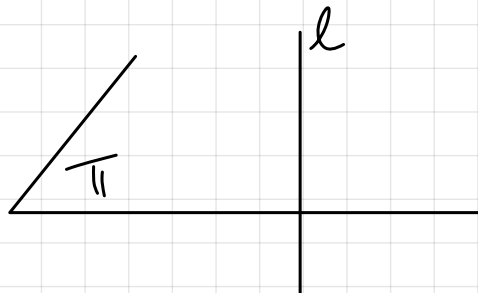
$$\begin{cases} 3x - 2y + z + 3 = 0 & |(-4) \\ 4x - 3y + 4z + 1 = 0 \end{cases} \quad (-)$$

$$\begin{cases} -8x + 5y - 11 = 0 \\ 4x - 3y + 4z + 1 = 0 \end{cases}$$

$$\Leftrightarrow \begin{cases} x = \frac{-5y + 11}{-8} \\ 4x - 3y + 4z + 1 = 0 \end{cases}$$

$$\Leftrightarrow \begin{cases} x = \frac{-5y + 11}{-8} \\ \frac{5y - 11}{2} - 3y + 4z + 1 = 0 \end{cases}$$

$$\Leftrightarrow \begin{cases} x = \frac{5y - 11}{8} \\ z = \left(3y - \frac{5y - 11}{2} - 1 \right) \cdot \frac{1}{4} \end{cases}$$



$$\Leftrightarrow \begin{cases} x = \frac{5y-11}{8} \\ z = \frac{y+9}{8} \end{cases} \Leftrightarrow \begin{cases} x = \frac{5\lambda-11}{8} \\ z = \frac{\lambda+9}{8} \\ y = 2 \end{cases}$$

$$D(\lambda) = \left\langle \left(\frac{5}{8}, 1, \frac{1}{8} \right) \right\rangle = \left\langle \underbrace{(5, 8, 1)}_{\vec{n}_{\Pi}} \right\rangle$$

$$\ell \perp \Pi \Leftrightarrow \vec{v} \parallel \vec{n}_{\Pi}$$

$$\vec{n}_{\Pi} = (a, 8, c) \Rightarrow \frac{a}{5} = \frac{8}{8} = \frac{c}{1} \Rightarrow \begin{cases} a=5 \\ c=1 \end{cases}$$